

ROSEHEARTY, United Kingdom

WMO: 030940

Lat: 57.700N

Lon: 2.117W

Elev: 7

StdP: 101.24

Time Zone: 0.00 (EUW)

Period: 95-12

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.1	0.1	-4.7	2.5	1.2	-3.6	2.8	1.7	23.2	5.6	21.1	5.8	4.3	170	0.794

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.0	20.6	16.7	19.0	15.6	17.7	14.8	17.2	19.8	16.2	18.3	15.4	17.1	5.8	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.1	11.4	18.4	15.2	10.8	17.3	14.5	10.3	16.3	48.4	19.9	45.5	18.4	43.1	17.1	21.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.2	15.5	13.7	-3.3	24.5	1.9	1.8	-4.6	25.8	-5.7	26.9	-6.8	27.9	-8.2	29.2	
			DB												
			WB	-4.3	19.4	2.0	1.3	-5.7	20.3	-6.9	21.1	-8.0	21.8	-9.5	22.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.0	4.8	5.1	5.9	7.6	9.3	11.9	14.0	14.1	12.8	10.2	7.2	4.9
	DBStd	4.06	2.02	2.37	2.24	2.47	2.16	2.19	2.07	1.90	1.97	1.98	2.43	2.55
	HDD10.0	821	160	139	129	79	40	6	0	0	2	22	87	159
	HDD18.3	3409	418	372	387	322	281	192	137	133	166	252	333	417
	CDD10.0	458	0	1	1	7	17	64	123	126	87	28	4	1
	CDD18.3	2	0	0	0	0	0	0	1	1	1	0	0	0
	CDH23.3	4	0	0	0	0	0	0	3	1	1	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	6.6	8.2	7.6	7.2	6.1	5.3	5.2	5.2	5.4	6.2	7.8	7.8	7.7
Precipitation	PrecAvg	851	77	62	54	58	58	68	71	72	66	99	92	74
	PrecMax	1044	223	111	110	138	131	127	132	158	267	200	179	163
	PrecMin	596	27	13	12	15	17	25	33	19	27	22	42	27
	PrecStd	107	41	29	21	34	26	31	32	33	47	40	37	28
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.8	13.1	13.4	16.5	18.4	21.2	22.7	22.6	21.5	16.8	14.3	12.9
		MCWB	9.9	10.3	10.3	13.0	14.0	17.0	17.3	17.9	17.1	14.1	11.7	10.8
	2%	DB	10.2	11.6	11.5	14.3	16.1	18.6	20.5	20.2	18.5	15.1	12.7	10.8
		MCWB	8.5	9.3	9.0	11.1	12.7	15.4	16.6	16.5	15.1	12.7	10.9	9.2
	5%	DB	8.9	10.1	10.2	12.7	14.3	16.8	18.9	18.4	16.9	14.0	11.7	9.4
		MCWB	7.5	8.3	8.1	10.1	11.5	14.1	15.6	15.5	14.2	12.3	10.3	8.0
	10%	DB	7.8	8.6	9.2	11.4	12.7	15.5	17.4	17.1	15.8	13.1	10.8	8.5
		MCWB	6.8	6.9	7.6	9.3	10.4	13.1	14.8	15.0	13.5	11.8	9.6	7.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.1	10.6	10.8	13.4	15.0	17.3	18.8	18.2	17.6	14.5	12.5	10.9
		MCDB	11.5	12.6	12.7	16.0	17.5	20.3	21.6	21.5	20.6	16.0	13.5	12.7
	2%	WB	8.8	9.5	9.5	11.6	13.1	15.7	17.2	17.1	15.9	13.4	11.4	9.3
		MCDB	9.9	11.4	10.9	13.5	15.3	18.1	19.7	19.5	17.9	14.4	12.4	10.5
	5%	WB	7.7	8.4	8.6	10.5	11.8	14.5	16.1	16.1	14.7	12.7	10.5	8.3
		MCDB	8.6	9.9	9.8	12.3	13.8	16.5	18.3	17.8	16.2	13.6	11.5	9.3
	10%	WB	6.9	7.2	7.7	9.5	10.8	13.5	15.1	15.3	14.0	12.1	9.8	7.3
		MCDB	7.8	8.4	8.9	11.0	12.3	15.1	16.9	16.5	15.2	13.0	10.7	8.3
Mean Daily Temperature Range	5% DB	MDBR	3.8	4.3	4.5	5.2	5.5	5.0	5.1	5.0	5.1	4.7	4.3	4.3
		MCDBR	5.3	6.3	6.4	7.6	8.4	8.1	8.2	7.6	7.0	6.0	5.1	5.4
		MCWBR	4.7	5.1	4.7	5.3	5.7	5.5	5.1	4.8	4.7	4.8	4.6	4.9
	5% WB	MCDBR	5.0	5.8	5.8	7.4	7.8	7.9	7.6	6.7	6.3	5.5	4.6	5.1
		MCWBR	4.6	5.1	4.6	5.5	5.5	5.6	5.2	4.6	4.7	4.7	4.4	4.8
Clear-Sky Solar Irradiance	taub	0.313	0.338	0.361	0.384	0.376	0.376	0.382	0.370	0.355	0.342	0.318	0.297	
	taud	2.237	2.322	2.333	2.298	2.379	2.409	2.401	2.441	2.481	2.420	2.277	2.170	
	Ebn at Noon	529	696	785	824	859	863	848	834	791	688	519	421	
	Edh at Noon	50	72	92	111	109	107	106	95	79	65	48	40	
All-Sky Solar Radiation	RadAvg	0.41	1.04	2.15	3.58	4.88	4.80	4.64	3.70	2.51	1.30	0.54	0.28	
	RadStd	0.05	0.10	0.21	0.33	0.27	0.39	0.35	0.24	0.21	0.11	0.04	0.03	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (26 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page